Python Fundamentals

30 Theory Questions

1. What is Python, and why is it called a high-level programming language?

2. Explain any five important features of Python.

3. What is the difference between a compiler and an interpreter?

4. Explain the internal working flow of a Python program from writing source code to execution.

5. What is PVM (Python Virtual Machine), and what role does it play in program execution?

6. What is a variable in Python, and how is a variable name associated with an object?

7. What does the id() function return in Python? Does it always return the same value for equal values?

8. What is the purpose of the type() function? Give examples of the kinds of information it returns.

9. What is the input() function in Python, and what type of value does it return by default?

10. Why do we often need to convert a value received through input() before performing calculations?

11. Explain the int, float, complex, str, and bool data types with one example of each.

12. What is the difference between an integer and a floating-point number in Python?

13. What is a complex number in Python, and how are its real and imaginary parts represented?

14. What are Boolean values in Python? Explain True and False and their common uses.

15. What is type casting (type conversion), and why is it used in Python?

16. Differentiate between implicit type conversion and explicit type casting.

17. What is an operator in Python, and what is an operand? Explain with a simple example.

18. What are the main categories of Python operators? Briefly describe each category.

19. What is the difference between arithmetic, relational, and logical operators?

20. What is the difference between the == operator and the is operator in Python?

21. What is a string in Python, and how can a string be declared using different quotation marks?

22. Why are Python strings immutable? What happens when we try to change one character of a string?

23. What is string indexing? Explain positive indexing and negative indexing.

24. What is string slicing, and what do start, stop, and step represent in a slice expression?

25. Why might a string slice return an empty string when the start, stop, and step values are incompatible?

26. What is the print() function, and how do the sep and end parameters affect its output?

27. What is the difference between print() and input() in Python?

28. What are identifiers and keywords in Python? State the rules for naming identifiers.

29. What is the difference between mutable and immutable objects in Python? Give examples.

30. How does Python determine the type of an expression, and how can type(), id(), and print() help us inspect a program while learning?